



3D rendering robotic arms with chipset for semiconductor manufacturing

witnessing automation's expansion into software development, challenging traditional processes and opening new avenues for efficiency. Tools like GitHub Copilot are revolutionising software development, enabling developers to work more productively.

This shift has profound implications for the workforce. For instance, companies like Accenture have begun re-evaluating their hiring strategies, opting to leverage these technologies rather than expanding their junior developer teams. This strategy underscores a broader trend towards automation that could reshape the economic landscape far sooner than anticipated. Research suggests that by 2030, 30% of tasks in the UK economy could be automated. Given the pace of technological advancement, this transformation may arrive even more swiftly, highlighting the urgent need for us to adapt and innovate alongside these emerging tools.

A 'big little' problem

The world of electronics and semiconductors is currently facing a paradoxically 'big little' problem. On one hand, the issue seems minor—after all, an AI chip or any semiconductor

chip for that matter hardly weighs a gram. Yet, this seemingly small component has become a significant nuisance due to its scarcity, posing a considerable challenge to the tech industry's growth and innovation.

Microsoft, in its recent annual report, highlighted this very issue, flagging the availability of chips, including GPUs and AI-specific chips, as a considerable risk to its investors. Despite the tech giant's efforts to advance and deploy AI in its offerings, the scarcity of these essential chips could potentially hamper its growth trajectory. This situation underscores a broader industry challenge: although production capacities are operating at full throttle, they are still unable to meet the soaring demand for these critical components.

The evolution of computing technology has always been closely tied to advancements in semiconductor technology. From the era when the central processing unit (CPU) was the cornerstone of computing, powering the rise of personal computers, we have seen technology evolve to meet the demands of increasingly complex tasks. CPUs were adept at handling basic compute tasks, logic,

and control functions. However, the advent of graphic-intensive applications necessitated the development of graphics processing units (GPUs) capable of processing complex images and 3D graphics efficiently.

Today, the focus has shifted towards artificial intelligence processing units (AIPUs), specialised chips designed to accelerate AI tasks. These chips are critical for the functioning and advancement of AI technologies, including those developed by companies like OpenAI. Sam Altman, co-founder of OpenAI, highlighted in a testimony to the US Senate the challenges posed by the scarcity of these chips. Despite the widespread adoption of AI technologies like ChatGPT, the inability to secure a steady supply of necessary semiconductors has led to scalability issues and performance bottlenecks for users.

The semiconductor shortage is more than just a supply chain issue; it is a bottleneck that affects the entire spectrum of technological innovation, from personal computing to advanced AI developments. As we navigate this 'big little' problem, the tech industry must explore new strategies to ensure the availability of these vital components, balancing the scales of supply and demand to fuel the next wave of technological advancement.

The AI boom in electronics

The intersection of AI and electronics design and manufacturing marks a pivotal era of technological synergy. We are witnessing an unprecedented acceleration in the AI chip market, with a staggering 50% compound annual growth rate (CAGR) from last year to the current year. This surge underscores the transformative impact AI is having on the electronics sector, propelling it into a new phase of innovation and growth.

In India, this momentum is further bolstered by a significant